



Faculty of Computing
Software Engineering - SE2030
Year 2 Semester 1 (2025)

Web-based Recruitment Company System

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1. Introduction

1.1 Project Overview:

LankaJobsHub.com is a web-based recruitment platform developed using Java, specifically designed to bridge the gap between job seekers and employers in Sri Lanka. This system aims to provide a structured, accessible, and centralized platform for job listings, applications, and candidate evaluations. It enables seamless interaction between job seekers and hiring managers through user-friendly interfaces, offering real-time access to job opportunities, application tracking, and profile management. Furthermore, it introduces administrative tools that ensure platform security, user role management, and system integrity.

1.2 Objectives:

- To develop a scalable and maintainable recruitment system tailored to Sri Lanka's job market.
- To empower job seekers with a platform that helps them discover and apply to relevant opportunities easily.
- To equip recruiters and hiring managers with efficient tools to manage postings and evaluate candidates.
- To ensure the system is secure, role-based, and adheres to core software engineering principles.

1.3 Target Users / Stakeholders:

- Job Seekers: Individuals actively looking for employment opportunities.
- Company Hiring Managers: Representatives from organizations managing job postings and candidate reviews.
- Recruitment Managers: Supervisors overseeing platform-wide hiring activities.
- HR Assistants: Staff involved in daily candidate communication and filtering.
- Client Relations Executives: Individuals responsible for company engagement and satisfaction.
- IT Officers / Admin: Personnel maintaining system security and functionality.

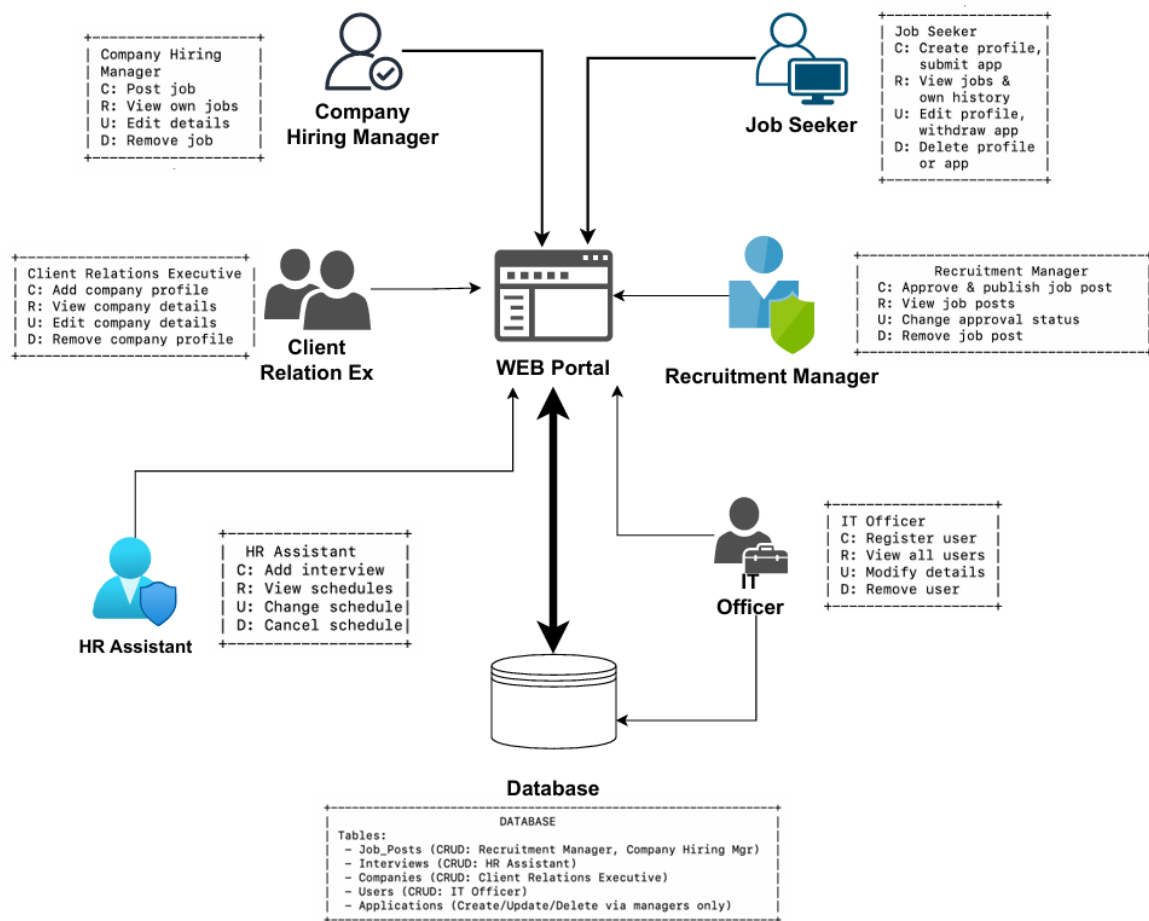
1.4 Scope:

The project covers a complete recruitment workflow, including account registration, job posting, resume submission, application filtering, and admin oversight. All data will be stored locally using file handling techniques in Java. Features such as job search with filters, notification prompts, and secure authentication are also within scope.

1.5 Limitations:

- The system will be limited to web access and optimized primarily for desktop browsers.
- No external API integrations (e.g., LinkedIn, email, SMS) are included in the initial version.
- Resume storage will be limited in file size and managed using local server directories.

2. System Overview Diagram



3. Functional Requirements

- User registration and authentication with role differentiation.
- Job posting interface for recruiters to add, edit, and delete listings.
- Profile creation and resume upload module for job seekers.
- Job search capability with filtering by category, location, and job type.
- Job application module with tracking and withdrawal functionality.
- Admin control panel for monitoring users, approving posts, and maintaining platform rules.

4. Non-Functional Requirements

- **Performance:** All main operations should execute within 2–3 seconds under normal load.
- **Security:** Passwords will be stored using encryption. Role-based access controls will restrict unauthorized access.
- **Usability:** The interface will be intuitive, responsive, and require minimal user training.
- **Maintainability:** The system will use modular Java classes and be well-documented to support future enhancements.
- **Scalability:** Although based on file handling, the design will support future migration to database-backed architecture.

5. Major Stakeholders / Users

- **Recruitment Manager:** Oversees all recruitment activities and supervises the overall system usage.
- **HR Assistant:** Assists with daily recruitment operations such as posting jobs and filtering applications.
- **IT Officer:** Handles backend maintenance, security updates, and technical troubleshooting.
- **Client Relations Executive:** Manages communication with hiring companies and ensures client satisfaction.
- **Job Seeker:** Interacts with the system to search and apply for jobs, and to maintain their digital resume.
- **Company Hiring Manager:** Creates job listings, views applications, and initiates interview processes.

6. Six Major Functions

1. Job Post Approvals

Assigned to: De Silva P.K.N (IT24102131)

User: Recruitment Manager

Function: Allows hiring managers to create, publish, update, or remove job listings.

Expected Outcome: Job listings are made visible to job seekers in real-time.

CRUD operations:

Create: Approve and publish a new job post.

Read: View pending, approved, and rejected job posts.

Update: Change approval status (e.g., approve → reject).

Delete: Remove an approved job post from listings

2. Candidate Interview Schedules

Assigned to: Kuruppuarachchige K.A.H.B (IT24102046)

User: HR Assistant

Function: Enables HR staff to search candidate profiles using filters such as skill, education, and location.

Expected Outcome: HR can efficiently generate lists of qualified applicants.

CRUD operations:

Create: Add new interview schedule for a candidate.

Read: View all upcoming and past interviews.

Update: Change interview date, time, or interviewer.

Delete: Cancel an interview schedule.

3. User Accounts

Assigned to: Withana W.Y.P (IT24102008)

User: IT Officer / Admin

Function: Admins manage access control, system logs, and enforce platform policies.

Expected Outcome: Secure platform operation and compliance with rules.

CRUD operations:

Create: Register a new system user account.

Read: View list of all system users.

Update: Modify user details (e.g., email, password).

Delete: Remove or deactivate a user account.

4. Employer Company Profiles

Assigned to: Pihara H.G.T (IT24102070)

User: Client Relations Executive

Function: Tracks employer account activity, job posting frequency, and candidate interactions.

Expected Outcome: Ensures active and satisfied employer participation.

CRUD operations:

Create: Add a new company profile to the system.

Read: View company details and status.

Update: Edit company information (e.g., contact person, address).

Delete: Remove a company profile from the database.

5. Profile & Applications

Assigned to: Dilshan R.M.R (IT24102181)

User: Job Seeker

Function: Enables users to upload resumes, enter personal details, and build a searchable profile.

Expected Outcome: Applications are recorded and accessible to recruiters for shortlisting

CRUD operations:

Create: Create new user profile, upload CV, submit job application.

Read: View job listings and application history

Update: Edit personal profile details or withdraw an application.

Delete: Remove profile or delete an application

6. Job Postings

Assigned to: Inshaf M.J.M (IT24100618)

User: Company Hiring Manager

Function: Creates, edits, and removes job postings they have published in the system.

Expected Outcome: Job listings remain accurate, updated, and relevant for potential candidates.

CRUD operations:

Create: Post a new job opening.

Read: View all job postings they created.

Update: Edit job posting details (e.g., job title, description)

Delete: Remove a job posting from the system.

Common Supportive Functions:

- Login/logout
- Password reset mechanism
- In-app notifications

7. Minor Functions

- Upload and update profile photos
- Bookmark jobs for future application
- Withdraw submitted applications
- Admin access to basic analytics dashboard

8. System Limitations / Constraints

- The system will be optimized for desktop browsers only, with minimal mobile support.
- No email, SMS, or third-party integration features in this phase.
- File size for resumes will be restricted to 2MB.
- All user and job data will be stored in structured text files using Java I/O.

9. Simple Timeline (Week 3 – Week 14)

Week	Milestone / Task
3	Project proposal, stakeholder identification
4	Use case design, initial wireframes
5	Backend setup (Java environment, file handlers)
6	User authentication module (register/login)
7	Job posting and profile creation modules
8	Application submission and tracking logic
9	Admin and user role integration
10	Search and filter functions
11	System testing and debugging
12	UI polishing and final integration
13	Report writing, documentation completion
14	Project presentation and viva preparation

10. Conclusion

LankaJobsHub.com is designed to meet the growing need for an efficient and accessible digital recruitment platform in Sri Lanka. The system aims to modernize traditional hiring methods by reducing dependency on paper-based processes and fragmented job postings. By offering an all-in-one platform for job seekers and employers, the system delivers real-world functionality and academic relevance.

Our development approach emphasizes modularity, secure Java-based implementation, and real-time interaction capabilities. With well-defined user roles and clearly assigned functions, each team member will contribute to the successful and timely delivery of the platform. This project not only aligns with the objectives of the SE2030 module but also reflects our commitment to practical software engineering practices. The system has significant potential for real-world application and can be further enhanced in future iterations.

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